MODEL IDENTIFICATION AND DATA ANALYSIS (Automation&Control Engineering), A.Y. 2020/2021

Prof. Sergio M. Savaresi, Prof. Simone Formentin – **July 13, 2021**

Surname	**Name**	**University ID Number**	**Signature**
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- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
- The number of pages is 8. Additional papers will not be considered.
- Use of books, lecture notes or any other material is forbidden. Electronics (smartphones, calculators, etc.) are forbidden.
- Clarity, order and precision are strongly considered for the final evaluation.

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1. [Supervised learning]

In the Italian SLAS factory, a predictive maintenance system must be set up to prevent faults and to react in time on the plant in case of need. The CEO of SLAS asks you to design it as a learning machine that estimates the probability of a fault and a model-based optimizer that decides how often to do the maintenance to minimize the risk of a fault.

a. Which hypothesis set could be appropriate for the problem at hand? Describe it in mathematical detail.
b. What data do you need to train such a model?
c. Describe the maximum likelihood approach that could be used to train the model.
d. List a couple of situations in which you expect to measure bad generalization performance.
e. How would you design the model-based decision-making system that sets the frequency of the maintenance?

SOLUTIONS:

a. The logistic model class, as a probability needs to be returned as an output. See the lecture notes (Chapter 04) for the mathematical details.
b. A set of operating conditions of the plant when a fault occurs, as well as a set of working conditions of the plant when this operates safely. The larger the dataset is, the better the performance is expected to be.
c. See the lecture notes (Chapter 04).
d. Bad generalization performance may occur in different scenarios, e.g., when the dataset is small as compared to the number of features considered in the model, or when the dataset is unbalanced (for instance, if the number of faults in the dataset is too low).
e. One possibility is to design it as a system that takes the probability of a fault as the input and returns a binary value: 1 when a maintenance intervention is needed and 0 otherwise. Such a system could be built as a simple logic block: if the probability of a fault is higher than a user-defined threshold, return 1, otherwise return 0.

2. [Unsupervised learning]

Consider the following tasks in unsupervised learning. Would you consider using PCA or K-means (or both)?

a. Data compression (namely, reduction of the necessary memory to save the dataset);

b. Grouping;

c. Data visualization;

d. Find structure in the data.

Provide adequate motivation to your answers. Then, briefly describe the working principles of the two methods.

SOLUTIONS:

a. PCA and K-means: if with PCA we can find the linear operator which minimizes the reconstruction error, with K-means we can consider as useful information the centroids and effectively reduce the dimension of the data.
b. K-means: the output of the K-means clustering algorithm is a set of labels to partition the original dataset into groups.
c. PCA: by considering only 2 or 3 principal components, we are able to visualize even datasets with high dimensionality.
d. PCA and K-means: in a different way, but both the algorithms can provide information about the structure of the data. In general, this could describe any unsupervised learning algorithm.

See Chapters 10 and 11 of the lecture notes for the description of the methods

3. [Stochastic processes]

Considering a zero mean stationary stochastic process $y(t)$.

a. Define its covariance function and explain why it turns out to be an even function.

b. Discuss how to estimate the mean and the covariance function of $y(t)$ based on a set of N measurements $\{y_1, y_2, \ldots, y_N\}$.

c. Suppose now that $y(t)$ is a white noise. Explain why it can be said that $y(t)$ is "unpredictable".

SOLUTIONS:

See Chapter 12 of the lecture notes for point a and c, and Chapter 13 for point b.

4. **[Fundamentals of learning]**

Explain why the following problems can or cannot be addressed by Machine Learning (ML) techniques:

a. Partitioning the students of this course;

b. Fortune-telling a person information about her/his personal life;

c. Determining the solution of a third-order equation;

d. Computing the vibrations induced on the driver of a car, given the physical models of vehicle, suspensions and road;

e. Provide weather forecasts.

In the case the problem can be addressed by ML, provide a suggestion for the technique you would use to solve the problem.

SOLUTIONS:

a. The problem of dividing into categories the students of this course can either be a machine learning problem (e.g., if we do not know the criterion used to partition) or not (e.g., if we are given the criteria). In the first case, we would use some unsupervised ML techniques (e.g., clustering) by considering for instance the personal data of each student.

b. In principle, fortune-telling is not science. If you consider it as a process where a fortune-teller is able to infer information about a person by looking at her/him, we could use a ML algorithm to determine the important features to infer information from a person picture (feature selection) and another algorithm able to couple the appropriate phrase to provide to each person (classification).

c. We have an equation and we simply want to derive its solution. It would be unnecessarily complicate to use a ML algorithm (whatever this would implies), since it exists a deterministic numerical procedure to provide the answer in a fast way.

d. The computation of the vibrations is simply the application of the mathematical or physical model to a specific case. Here, we might resort to analytical solutions, in the case they exist, or numerical techniques in the field of scientific computing.

e. In this case, we are considering a regression problem. One might consider the use of static models, if we take into account the weather of each day as independent from the previous days. If we want to take into account the fact that the temperatures and pressures affecting the weather are time series (there exists correlation among days), we should resort to time series prediction.

5. [Analysis and prediction of ARMA/ARMAX processes]

Consider the process $y(t)$, defined as

$$y(t) = \frac{1}{3}y(t-1) + 3e(t) + e(t-1)$$

with $e \sim wn(1,1)$.

a. What kind of process is this?

b. Is the process in canonical form? If the process is not in canonical form, bring it in canonical form.

c. Is $y(t)$ stationary?

d. Compute the mean m_y of the process.

e. Compute the optimal 1-step ahead predictor $\hat{y}(t|t-1)$.

SOLUTIONS:

a. The process is an ARMA(1,1).

b. The process is not in canonical form, since C(z) is not monic. To bring it in canonical form we have to introduce $\xi(t) = 3e(t)$, with $\xi \sim wn(3,9)$. The transfer function (in canonical form) characterizing the process thus becomes:

$$y(t) = \frac{1+\frac{1}{3}z^{-1}}{1-\frac{1}{3}z^{-1}}$$

c. The process is stationary since the roots of A(z) are inside the unit circle.

d. The mean of the process is given by $m_y = \frac{1}{3}m_y + 4$, thus resulting in $m_y = 6$.

e. To find the one step ahead predictor, we initially detrend the output and the noise obtaining:

$$\tilde{y}(t) = y(t) - 6, \qquad \tilde{\xi}(t) = \xi(t) - 3.$$

The evolution of the zero-mean signal $\tilde{y}(t)$ is derived from the one of the original process, and it is given by

$$\tilde{y}(t) = \frac{1}{3}\tilde{y}(t-1) + \tilde{\xi}(t) + \frac{1}{3}\,\tilde{\xi}(t-1).$$

We can thus compute the optimal one step ahead for the detrended signal, i.e.,

$$\hat{\tilde{y}}(t|t-1) = -\frac{1}{3}\,\hat{\tilde{y}}(t-1|t-2) + \frac{2}{3}\,\tilde{y}(t-1)$$

By reintroducing the mean and exploiting the DC gain of the predictor, the final one-step ahead predictor is

$$\hat{y}(t|t-1) = -\frac{1}{3}\,\hat{y}(t-1|t-2) + \frac{2}{3}y(t-1) + 3$$

6. **[State-space identification]**

Given the following samples of the impulse response of a system

$\omega(0) = 0, \omega(1) = 3, \omega(2) = 1, \omega(3) = \frac{1}{3}, \omega(4) = \frac{1}{9}, \omega(5) = \frac{1}{27}$

a. Identify the order of the system.
b. Identify the state space model of the system with the 4SID method.
c. Find the transfer function of the system.
d. Find the corresponding state-space representation in control form.
e. Construct H_3 from the matrices of the state space model in control form and check that $H_3 = O_3 R_3$.

SOLUTIONS:

a. The Hankel matrix of the system is $H_1 = 3$, and thus it has unitary rank. The matrix $H_2 = [\,3, 1; 1, \frac{1}{3}]$ is not full rank since the first row is equal to three times the second one. Therefore, $rank(H_2) = 1$ and thus $n = 1$.

b. $H_2 = O_2 R_2$, where R_2 is selected as the first row of the Hankel matrix H_2, namely $R_2 = [3\ 1]$, while $O_2 = [1; \frac{1}{3}]$ is selected to satisfy the equality. This allows us to retrieve the matrices of the state space model as follows:

$$\widehat{H} = O_2(1) = 1,\ \ \widehat{G} = R_2(1) = 3, \widehat{F} = O_2(1)^{-1} O_2(2) = \frac{1}{3}, \widehat{D} = 0$$

and, thus, the model of the system retrieved with 4SID is

$$x(t+1) = \frac{1}{3} x(t) + 3u(t)$$
$$y(t) = x(t)$$

c. The transfer function associated with the previously obtained state space model is $W(z) = \widehat{H}\left(z - \widehat{F}\right)^{-1} \widehat{G} = \frac{3}{z - \frac{1}{3}}$

d. The matrix of the state space model in control form are $\overline{F} = \frac{1}{3}, \bar{G} = 1, \overline{H} = 3, \overline{D} = 0$.

e. The Hankel matrix $H_3 = [3\ 1\ \frac{1}{3}; 1\ \frac{1}{3}\ \frac{1}{9}; \frac{1}{3}\ \frac{1}{9}\ \frac{1}{27}]$. The observability matrix is given by $O_2 = [\overline{H}; \overline{H}\ \overline{F}; \overline{H}\overline{F}^2] = [3; 1; \frac{1}{3}]$, while the reachability matrix is given by $R_3 = [\bar{G}\ \ \overline{F}\bar{G}\ \ \overline{F}^2\bar{G}] = [1\ \frac{1}{3}\ \frac{1}{9}]$. Computing the product between these two matrices, the equality is easily proven.

7. **[Kalman filter]**

Assume that there exist an asymptotic solution of the Kalman filter, with asymptotic gain $\overline{K}$.
Find the condition of asymptotic stability for the asymptotic Kalman Filter (the proof of this condition is requested)

SOLUTIONS:
See lecture notes

8. [SW-sensing]

Compare model-based SW-sensing (Kalman Filter) and Black-box SW-sensing. List the main 6 comparison features and highlight the pros and cons of the two methods, according to such features.

SOLUTIONS:

See lecture notes